



ITSimplera Solutions

Exploratory Data Analysis Report

Online Retail II Dataset — Week 1 Task

Artificial Intelligence and Machine Learning Internship

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Executive Summary

The report analyses 525,461 transaction records from a UK-based online retailer to prepare the ground for a future product recommendation system. The analysis is observation-only: nothing has been deleted, corrected, or filtered out of the dataset. Instead, every pattern and every quality issue found is reported so that the business can make deliberate, informed decisions about how to treat the data going forward.

Three findings matter most. First, the business is highly concentrated one country supplies 85.9% of revenue, and a small group of 'hero' products drives a large share of units sold. Second, the data itself has real gaps: a fifth of all rows cannot be tied to a customer, and roughly 2% of invoices are cancellations sitting inside the raw totals. Third, the customer base already shows a strong repeat purchase habit, which is exactly the signal a recommendation engine needs to work well. Beyond the core requirements, this report also includes an extended analysis a Pareto (80/20) check, a day-of-week/hour demand pattern, a gross vs. net revenue comparison, and an RFM customer segmentation to give the business a more complete starting picture.

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1. Introduction and Business Context

The report documents the first-week task of the internship: a full exploratory data analysis (EDA) of a real e-commerce transaction dataset. The scenario given by the company is that of a retail business that is considering building a product recommendation system. Before any modeling work can begin, the business needs a clear picture of its own data how customers behave, which products actually sell, which countries the revenue comes from, and where the dataset itself has gaps or quality problems that could mislead later analysis.

The work in this report is deliberately observation-only. No rows have been deleted, filtered, or corrected at this stage. Every issue found in the data missing customer IDs, cancelled orders, duplicate rows, negative values is reported here so that the business can make an informed decision on how to treat it later, rather than having that decision made silently during cleaning.

2. Dataset Overview

Attribute	Detail
Source	UCI Machine Learning Repository
Coverage period	2009 – 2011, UK-based online retailer
Rows	525,461
Columns	8 (InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country)
Grain	One row per product line item within an invoice

Quantity and UnitPrice are numeric, CustomerID is stored as a decimal number because of missing values, and the remaining fields are text. InvoiceDate is stored as text in the raw file and was converted to an actual datetime type before any time-based analysis.

3. Tools and Environment

- **Platform:** Google Colab (cloud Jupyter notebook)
- **Language:** Python 3
- **Libraries:** pandas, numpy, matplotlib, seaborn

4. Methodology

Step 1: Loading the Data and Basic Structure

The CSV file was read with pandas using utf-8-sig encoding, which strips a hidden byte-order-mark character that would otherwise attach itself to the first column name. The shape, column list, data types, and a preview of the first rows were printed to confirm the file loaded correctly before any further work began.

Step 2: Missing Values and Duplicate Rows

Every column was checked for missing values using `isnull()`, and the counts were converted into percentages so the scale of the problem is easy to read at a glance. Two columns carry almost all of the missing data:

Column	Missing Rows	Missing %
CustomerID	107,927	20.54%
Description	2,928	0.56%

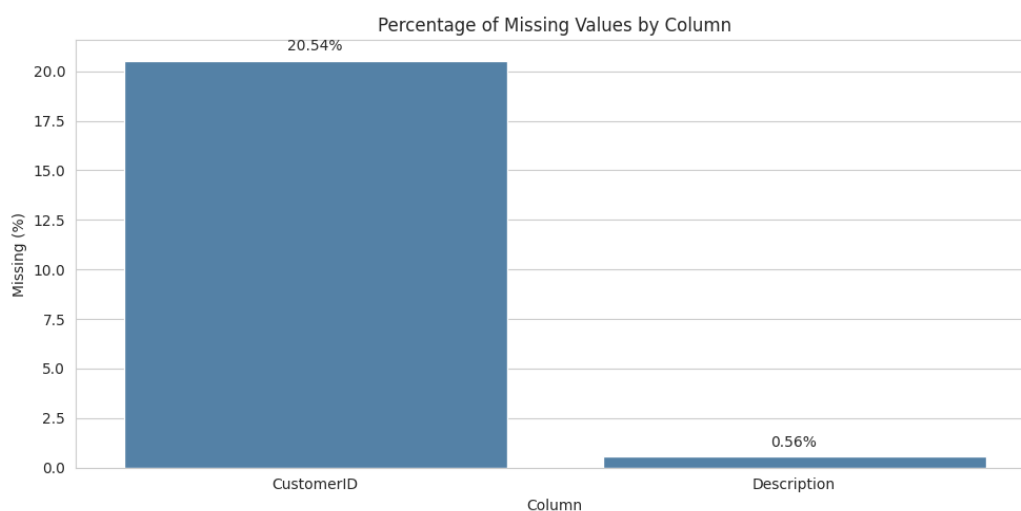


Figure 1 — Missing values by column

A check for fully duplicated rows (every column identical) found 6,865 rows, roughly 1.31% of the dataset. These were counted, not removed, in line with the task instructions.

Step 3: Additional Data Quality Checks

Beyond the checks explicitly asked for, a few extra checks were run because they change how the rest of the numbers should be read. In this dataset, any InvoiceNo that starts with the letter 'C' marks a cancelled or returned order rather than a genuine sale.

Check	Result
Cancelled invoices (InvoiceNo starts with 'C')	10,206 rows (1.94%)
Negative Quantity	12,326 rows
Negative UnitPrice	3 rows
UnitPrice equal to 0	3,687 rows
Missing CustomerID	107,927 rows (20.54%)

None of these rows were removed. They stay in every chart and total that follows, which is why the revenue and quantity figures below should be read as raw totals rather than 'clean sales' figures.

Step 4: Derived Columns for Analysis

Three columns were added on top of the original data, without changing or deleting anything that was already there:

- **Revenue:** Line-item revenue, calculated as Quantity multiplied by UnitPrice.
- **InvoiceMonth:** The invoice date reduced to a year-month label, used to group sales by month.
- **IsCancelled:** A true/false flag marking whether the invoice number starts with 'C'.

Step 5: Top 10 Best-Selling Products

Products were ranked twice — once by total units sold and once by total revenue generated because the two lists do not overlap much. The best-selling item by volume, the White Hanging Heart T-Light Holder, is a low-price, high-frequency product, while the top item by revenue, the Regency Cakestand 3 Tier, earns its place through a higher price point rather than sheer order count.

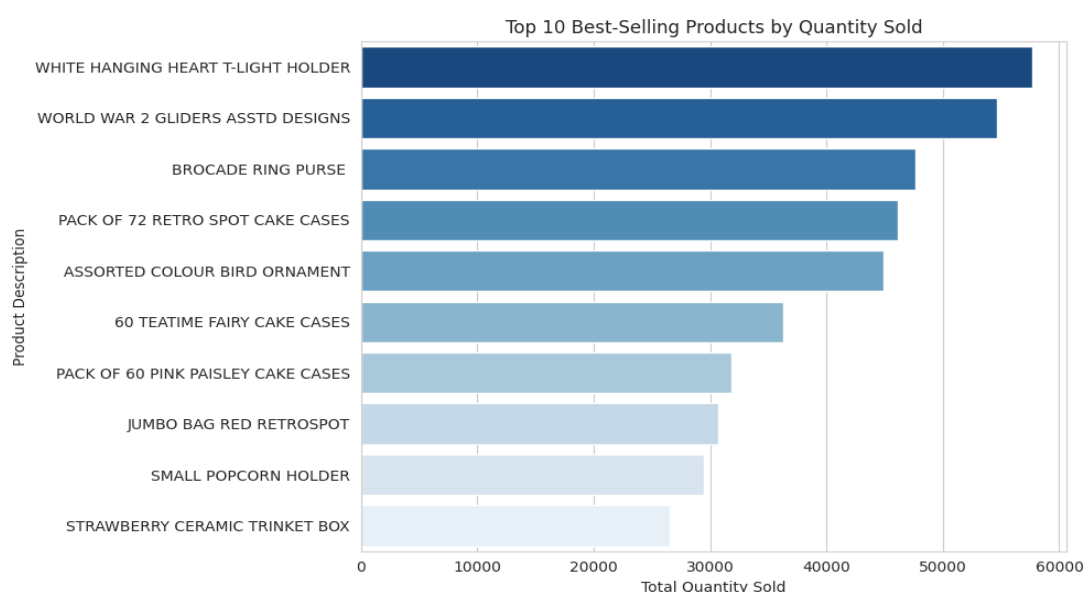


Figure 2 — Top 10 products by quantity sold

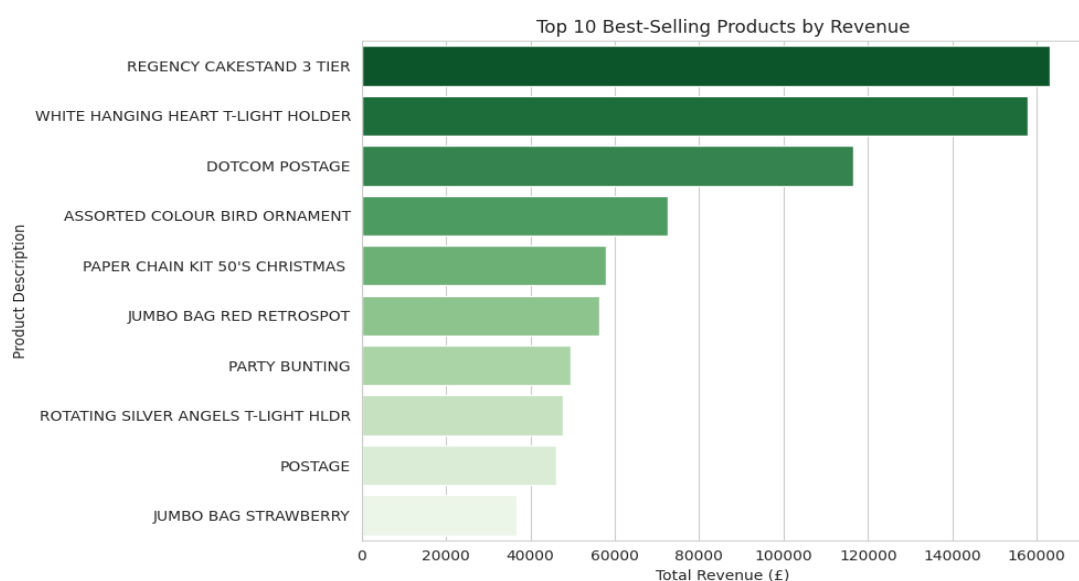


Figure 3 — Top 10 products by revenue (£163,051.46 for the top item)

Step 6: Sales Performance by Country

Revenue, unique invoice count, total quantity, and unique customer count were aggregated per country. Because the United Kingdom accounts for the large majority of the business, a second chart with the UK excluded was produced so the remaining markets are actually visible on the same scale.

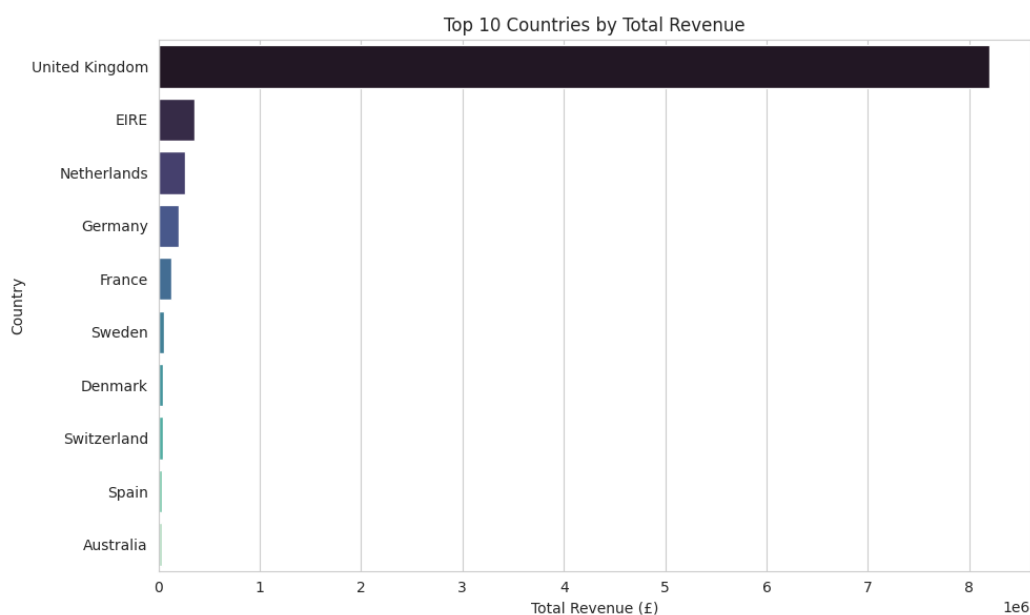


Figure 4 — Top 10 countries by total revenue

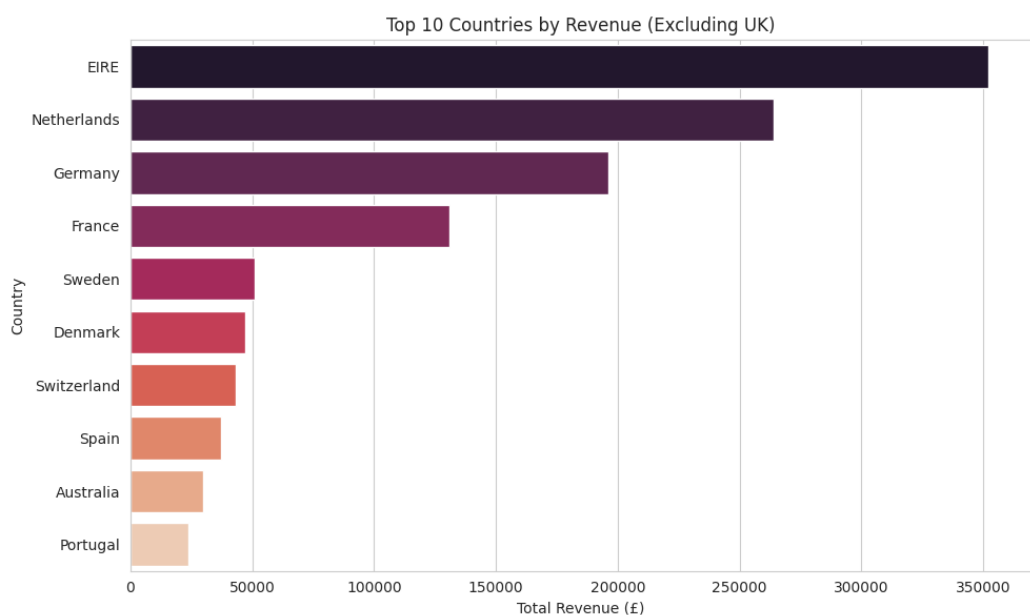


Figure 5 — Top 10 countries by revenue, UK excluded

The United Kingdom alone contributes 85.9% of total revenue. EIRE (Ireland), the Netherlands, Germany, and France form the next tier of markets.

Step 7: Revenue Over Time

Monthly revenue was aggregated and plotted as a line chart to look for seasonality. A second chart overlays monthly revenue with the number of unique orders per month, to check whether revenue swings are driven by more orders or by larger orders.

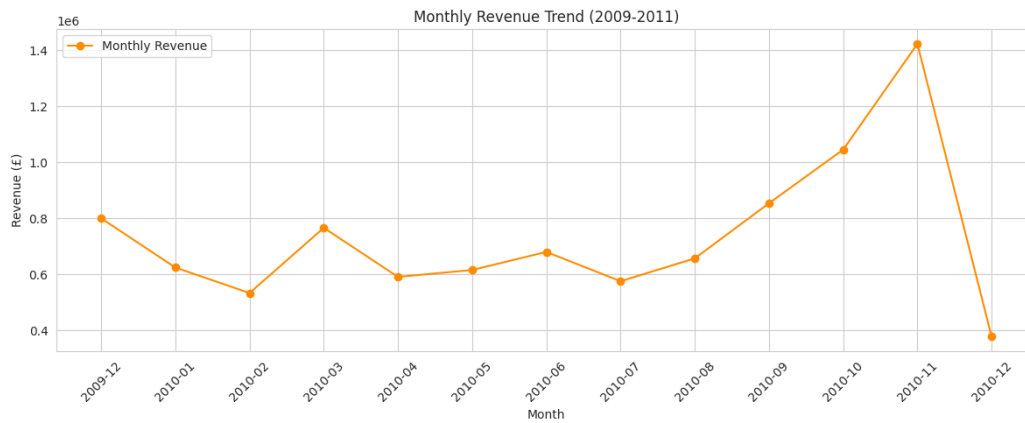


Figure 6 — Monthly revenue trend, December 2009 to December 2010

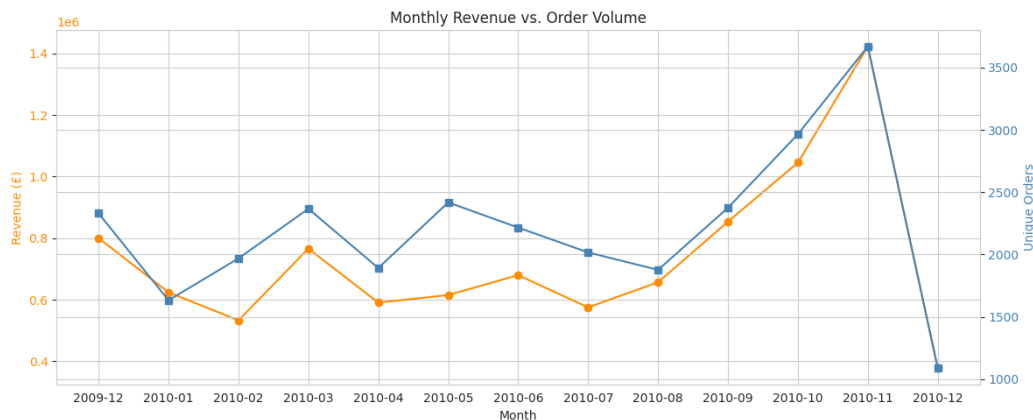


Figure 7 — Monthly revenue against monthly order volume

Revenue climbs steadily through the year and peaks sharply in November at £1,422,654.64, consistent with pre-holiday buying, before falling in December — which in this dataset reflects the point where the recorded data ends rather than a genuine seasonal drop.

Step 8: Correlation Heatmap

A Pearson correlation matrix was computed for the four numeric columns — Quantity, UnitPrice, Revenue, and CustomerID — and displayed as a heatmap.

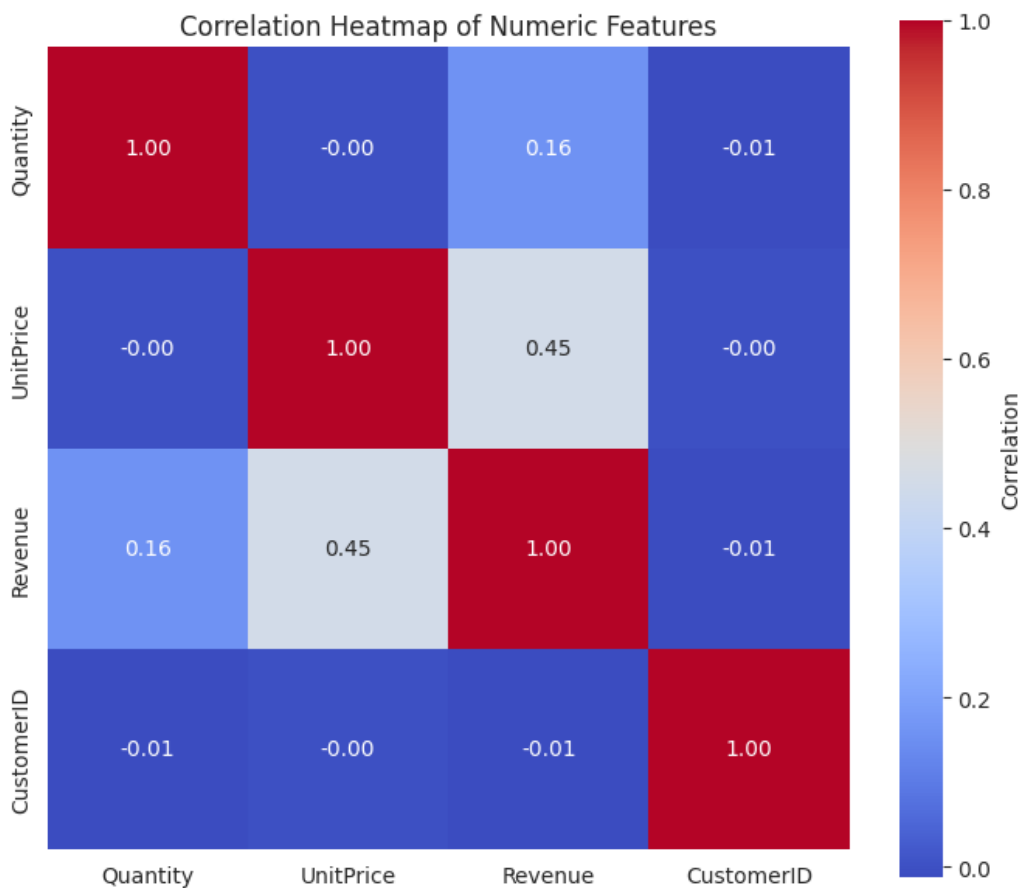


Figure 8 — Correlation heatmap of numeric features

Revenue shows a moderate positive correlation with UnitPrice (0.45) and a weak one with Quantity (0.16). In practice this means revenue behaves as a joint outcome of price and volume together, and neither column on its own is a reliable stand-in for revenue.

Step 9: Outlier Detection

Box plots were drawn for Quantity, UnitPrice, and Revenue, and outlier counts were computed separately using the interquartile range (IQR) rule, where any point more than 1.5 times the IQR beyond the first or third quartile is flagged.

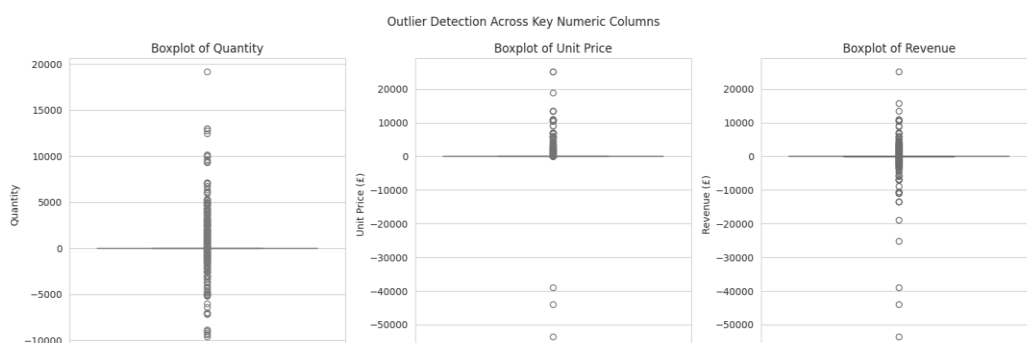


Figure 9 — Box plots of Quantity, UnitPrice, and Revenue

Column	Outlier Count	% of Rows
Quantity	57,870	11.01%
UnitPrice	35,273	6.71%
Revenue	45,872	8.73%

Step 10: Customer Behaviour Snapshot

As an extra check relevant to the recommendation-system goal, the number of distinct orders per identified customer was calculated.

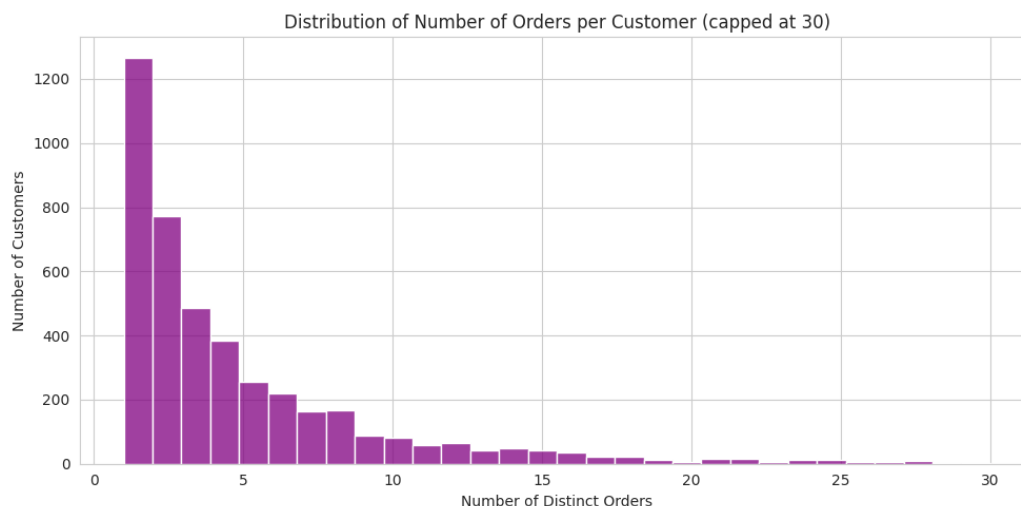


Figure 10 — Distribution of number of orders per customer (capped at 30 for readability)

Of the 4,383 customers who can be identified by a CustomerID, 3,116 — about 71.1% — placed more than one order. This repeat-purchase base is exactly the kind of purchase history a recommendation system relies on.

5. Extended Analysis

The five steps below go beyond the task's core requirements. They were added because they answer the questions a stakeholder would naturally ask next, and because they build directly toward the company's stated goal of a recommendation system. Every step below still follows the same rule as the rest of this report: nothing is deleted from the dataset, only computed on top of it. The corresponding code has been added to the notebook (Bonus Steps A–E); the notebook should be re-run end-to-end and the resulting charts inserted into this section before final submission.

Bonus Step A: Extended Statistical Summary

Beyond the boxplots in Step 9, a full statistical summary (mean, quartiles, skewness, kurtosis) is computed for Quantity, UnitPrice, and Revenue. Skewness tells us how lopsided a distribution is, and kurtosis tells us how heavy its tails are together they explain, numerically, why the boxplots show so many points above the upper whisker rather than being a coincidence of the chart type.

Bonus Step B: Pareto (80/20) Analysis

This step tests the classic Pareto principle directly: what share of total revenue comes from the top 20% of products, and separately, from the top 20% of identified customers? The result is plotted as a

cumulative Pareto curve with the 80% line and the 20% line marked, so it is immediately visible whether and how strongly the rule holds for this business.

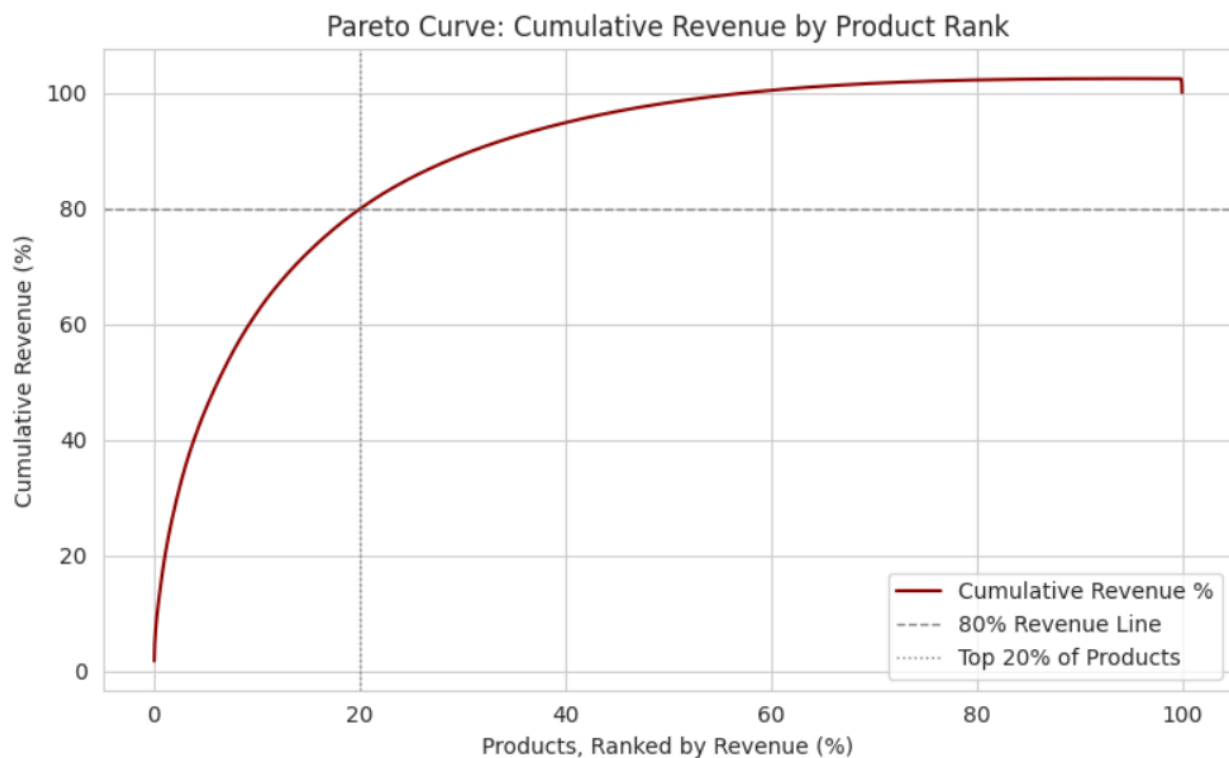
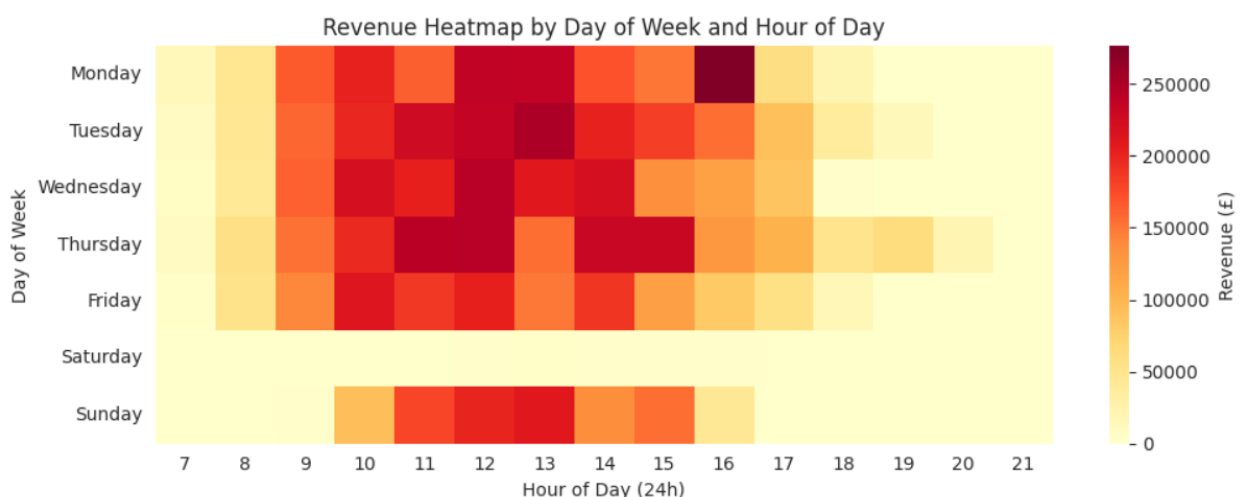


Figure 11 — Pareto (80/20) cumulative revenue curve

Bonus Step C: Day-of-Week and Hour-of-Day Demand Pattern

A heatmap of revenue by day of week and hour of day shows exactly when customers place orders, not just what they buy. This is directly useful for planning flash sales, customer support staffing, and server capacity questions that a purely product-level or country-level analysis cannot answer.



Highest-revenue day of the week: Thursday

Figure 12 — Revenue by day of week and hour of day

Bonus Step D: Gross vs. Net Revenue

Since cancelled invoices are still included in every revenue figure reported in Steps 6 and 7, this step quantifies the gap directly: gross revenue (everything) is compared side-by-side with net revenue

(cancellations excluded), so the business can see exactly how much the raw top-line number is inflated by returns.

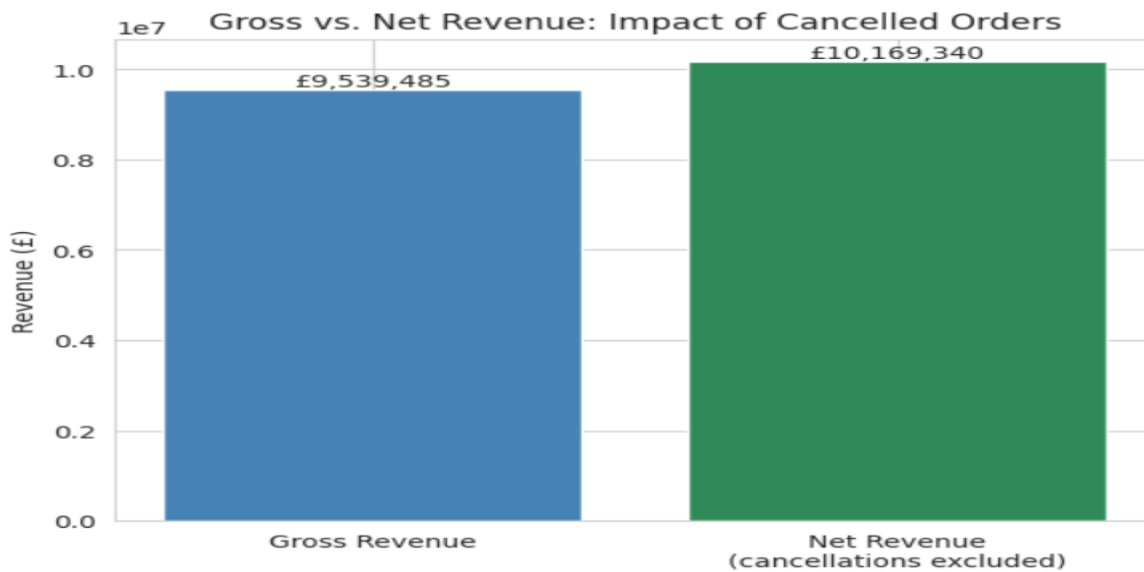


Figure 13 — Gross vs. net revenue comparison (cancellations excluded)

Bonus Step E: RFM Snapshot and Customer Segmentation

Recency, Frequency, and Monetary (RFM) value are calculated for every identified customer and combined into a single score. Customers are then grouped into four rule-based segments — Champions, Loyal, At Risk, and Needs Attention — using quartile thresholds only, with no machine learning model involved. This keeps the step within the spirit of an EDA task while giving the business its first concrete view of which customers matter most, which is normally the very first table a recommendation system is built on.

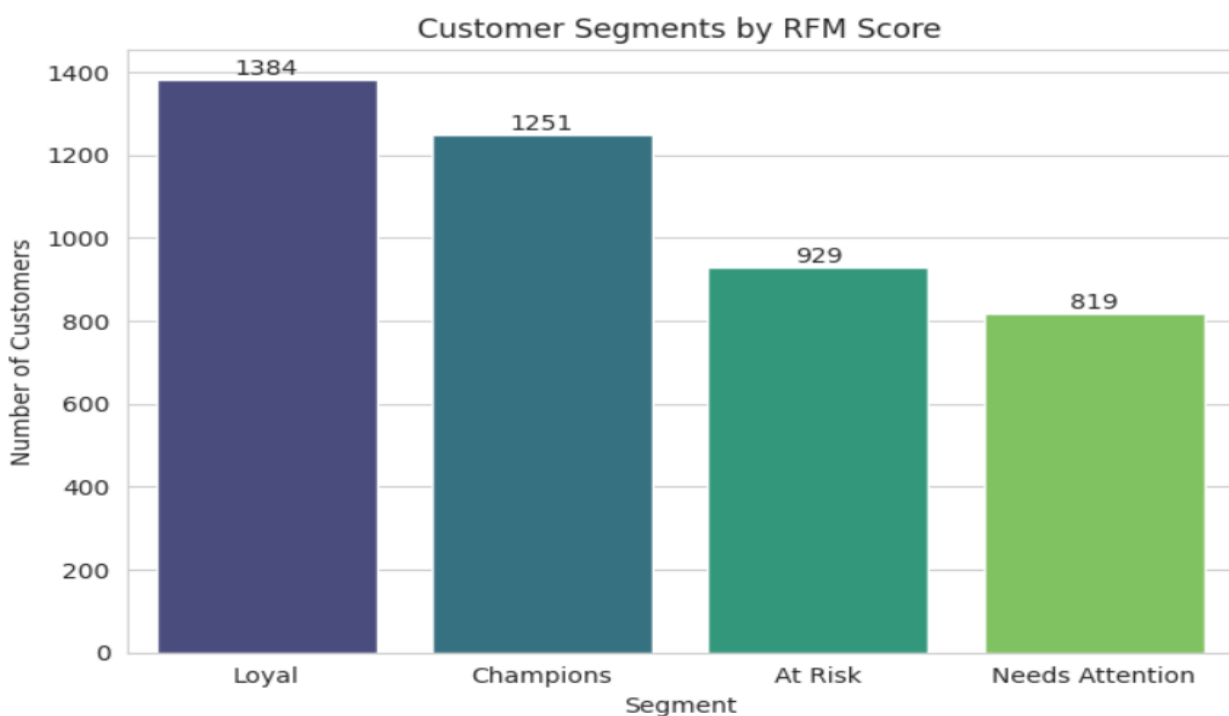


Figure 14 — RFM customer segmentation (Champions, Loyal, At Risk, Needs Attention)

6. Business Insights

The following insights are drawn directly from the analysis above. Each one is written as something the business can act on, not just a restated statistic.

1. The business is almost entirely UK-driven, with room to grow abroad.

The UK supplies 85.9% of total revenue. EIRE, the Netherlands, Germany, and France are already generating meaningful revenue as the next four countries in the ranking, which suggests these markets could absorb more marketing and logistics investment before the company looks any further afield.

2. One in five transaction rows cannot be tied to a customer.

CustomerID is missing on 107,927 rows (20.54% of the dataset). Any personalized recommendation system will simply have no history to work with for this share of transactions, so the business needs either a plan to capture customer identity more consistently at checkout, or a fallback strategy such as popularity-based recommendations for anonymous purchases.

3. Cancelled orders are currently mixed in with genuine sales.

10,206 rows (1.94%) are cancellations, identifiable because their invoice number starts with 'C'. Left inside the raw totals, they slightly inflate figures like 'units sold' and 'top revenue product'. Before these numbers are used as official KPIs, cancellations should be separated out.

4. A small set of 'hero' products carries a disproportionate share of volume.

The top 10 products by quantity sold represent a small fraction of the full product catalogue but a large share of total units moved. These products are strong candidates for guaranteed stock availability and for seeding a 'customers also bought' feature.

5. Sales are seasonal, with a clear November peak.

Monthly revenue builds steadily across the year and peaks at £1,422,654.64 in November, in line with pre-holiday shopping behaviour. Staffing, inventory, and marketing spend would benefit from being weighted toward the final quarter rather than spread evenly.

6. Most identified customers already return for a second purchase.

71.1% of the 4,383 identifiable customers have placed more than one order. This is a healthy signal for a future recommendation engine, since there is enough repeat-purchase history to personalize suggestions for the majority of known customers. First-time and anonymous shoppers will still need a separate 'cold start' strategy based on overall popularity.

7. Revenue cannot be predicted well from price or quantity alone.

The correlation heatmap shows only a moderate link between Revenue and UnitPrice (0.45), and a weak one between Revenue and Quantity (0.16). Any future model that tries to estimate revenue will need to combine both variables rather than lean on a single one.

8. A meaningful share of 'outliers' are probably bulk or wholesale orders, not data errors.

Between 6.7% and 11% of rows in Quantity, UnitPrice, and Revenue fall outside the normal IQR range. Given the size and consistency of this group, it more likely represents a distinct wholesale or

B2B customer segment than random data entry mistakes, and may be worth analysing separately rather than discarding.

7. Limitations and Assumptions

- The dataset only covers a UK-based retailer's transactions from 2009–2011; findings on seasonality and country mix should not be assumed to generalize to other markets or time periods.
- Revenue figures throughout Steps 1–10 are gross figures and include cancelled orders, since no rows were removed at this stage. Bonus Step D quantifies this gap separately.
- Customer-level analysis (repeat-purchase rate, RFM segmentation) only covers the 79.5% of rows that have a CustomerID; the remaining rows are, by definition, invisible to any customer-level view.
- The RFM segmentation in Bonus Step E is a rule-based quartile split, not a trained clustering model, and is intended as a first-pass view rather than a final segmentation strategy.

8. Recommendations and Next Steps

This report is deliberately observation-only no rows were removed at any point, as required by the task. The following steps are suggested as the natural next phase, before any modelling work begins:

- Decide how to handle the 107,927 rows with a missing CustomerID drop them from personalization, impute where possible, or route them to a popularity-based fallback.
- Separate the 10,206 cancelled invoices from genuine sales in all future revenue and best-seller calculations.
- Investigate the small number of negative and zero UnitPrice rows, which likely represent manual adjustments rather than real sales.
- Treat the detected outliers as a possible wholesale/B2B segment rather than removing them by default.

9. Conclusion

This EDA gives the business a grounded, evidence-based picture of its transaction data ahead of any recommendation-system work: who the customers are, which products and countries drive revenue, where the seasonal peaks fall, and just as importantly where the data itself is incomplete or noisy. Every finding here is reported rather than corrected, so that decisions about how to treat missing values, cancellations, and outliers remain visible and deliberate rather than buried inside a cleaning script. The extended analysis adds a first concrete step toward the company's ultimate goal a working set of customer segments and demand patterns that a recommendation system can be built on top of.

10. References

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